EVERY MORPHISM FACTORS THROUGH AN ADMISSIBLE MONOMORPHISM WITH A HOMOTOPY RETRACTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the direct-sum factorization object with its splittings, the assembly of tilde-alpha from the cone sequence, the admissible monomorphism verification with pi tilde-alpha = alpha and pi s = 1_y, and the explicit homotopy computation via lemma-id-cone-null are all retained verbatim.

Editorial note. The label lemma-factor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Axiom (A): direct sums in the differential graded category. Same chapter, Situation situation-ABC, axiom (A) and Lemma lemma-id-cone-null. Provides the differential graded direct sum y plus C(1_x) with its split projections and sections, and the lemma identifying the cone of the identity with a null object whose structure maps give the homotopy input.
- The cone sequence of the identity. Same chapter, Definition definition-cone and the admissible short exact sequence x to C(1_x) to x[1]. Supplies the natural morphism x to C(1_x) fitting the admissible sequence, out of which the admissible monomorphism tilde-alpha is assembled.

Source reference. Original reference 09QK: sources/stacks-dga/pinned-original-source.tex, source lines 3867–3867. Prerequisite pointer invoked at proof line(s) 4011; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09QJ: sources/stacks-dga/pinned-original-source.tex, source lines 3762–3762. Prerequisite pointer invoked at proof line(s) 3939; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09QJ let $\alpha : x \to y$ be a morphism in Comp($\mathcal{A}$). Then there exists a factorization in Comp($\mathcal{A}$):*

$$x \xrightarrow{\tilde{\alpha}} \tilde{y} \underset{s}{\overset{\pi}{\rightleftarrows}} y$$

such that

(1) *$\tilde{\alpha}$ is an admissible monomorphism, and $\pi\tilde{\alpha} = \alpha$.*

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(2) *There exists a morphism $s : y \to \tilde{y}$ in $Comp(\mathcal{A})$ such that $\pi s = 1_y$ and $s\pi$ is homotopic to $1_{\tilde{y}}$.*

Proof. By axiom (A), we may let $\tilde{y}$ be the differential graded direct sum of y and $C(1_x)$, i.e., there exists a diagram

$$y \underset{\pi}{\overset{s}{\rightleftarrows}} y \oplus C(1_x) \underset{t}{\overset{p}{\rightleftarrows}} C(1_x)$$

where all morphisms are of degree zero, and in $\mathrm{Comp}(\mathcal{A})$. Let $\tilde{y} = y \oplus C(1_x)$. Then $1_{\tilde{y}} = s\pi + tp$. Consider now the diagram

$$x \xrightarrow{\tilde{\alpha}} \tilde{y} \underset{s}{\overset{\pi}{\rightleftarrows}} y$$

where $\tilde{\alpha}$ is induced by the morphism $x \xrightarrow{\alpha} y$ and the natural morphism $x \to C(1_x)$ fitting in the admissible short exact sequence

$$x \rightleftarrows C(1_x) \rightleftarrows x[1]$$

So the morphism $C(1_x) \to x$ of degree 0 in this diagram, together with the zero morphism $y \to x$, induces a degree-0 morphism $\beta : \tilde{y} \to x$. Then $\tilde{\alpha}$ is an admissible monomorphism since it fits into the admissible short exact sequence

$$x \xrightarrow{\tilde{\alpha}} \tilde{y} \longrightarrow x[1]$$

Furthermore, $\pi\tilde{\alpha} = \alpha$ by the construction of $\tilde{\alpha}$, and $\pi s = 1_y$ by the first diagram. It remains to show that $s\pi$ is homotopic to $1_{\tilde{y}}$. Write 1_x as $d(h)$ for some degree -1 map. Then, our last statement follows from

$$\begin{aligned} 1_{\tilde{y}} - s\pi =& tp \\ =& t(dh)p \quad \text{(by Lemma 09QK)} \\ =& d(thp) \end{aligned}$$

since $dt = dp = 0$, and t is of degree zero. □